DAY-3 CONFIGURING BASIC NETWORK HARDWARE

Topic: IP Addressing Basics and Basic Network Configuration

Course: Diploma in Cyber Security

NAME:HET ACHARYA

Task 1: Login to Router Admin Panel

For this task, I first connected my laptop to my home Wi-Fi network.

I opened Command Prompt and used the following command:

ipconfig

I checked the Default Gateway shown under the Wi-Fi adapter. This address is normally the router's local IP address.

I then entered the Default Gateway address in a web browser to open the router's administration/login page.

Steps

1. Connect the laptop to Wi-Fi.

2. Open Command Prompt.
3. Type ipconfig.
4. Find Default Gateway.
5. Enter the gateway address in the browser.
6. The router login page opens.
7. Enter the router's login details if required.

Screenshot

Figure 1: Router Admin Panel Login Page

Note: The router IP can be different depending on the router model and network configuration. Common examples are 192.168.0.1 and 192.168.1.1.

Figure 1: Jio Router Admin Login Page

TASK 2 – CHECK IP ADDRESS DETAILS

Practical Task

For this task, I checked the network information of my laptop while it was connected to Wi-Fi.

I opened **Command Prompt** and used the following command:

ipconfig

From the result, I checked these three details:

IPv4 Address: 192.168.29.52

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.29.1

```
C:\Users\br-16>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2405:201:2012:4048:fde4:8e5d:ea0f:2282
   Temporary IPv6 Address. . . . . . : 2405:201:2012:4048:976:c01c:d480:dbce
   Link-local IPv6 Address . . . . . : fe80::b219:8087:954e:af3%7
   IPv4 Address. . . . . . . . . . . : 192.168.29.52
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::16ae:85ff:feea:2770%7
                                       192.168.29.1

C:\Users\br-16>
```

The **IP address** identifies my laptop on the network. The **subnet mask** shows the network range, and the **default gateway** is used to communicate with other networks and access the internet.

TASK 3 – STATIC IP ADDRESS

Practical Task

In this task, I changed my laptop's IP address from **automatic (DHCP)** to a **static IP address** in the same network range as my router.

My current network details are:

- **IPv4 Address:** 192.168.29.52
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.29.1

Static IP Configuration

For the static IP, I can use an unused IP address in the same network range, for example:

- **IP Address:** 192.168.29.50
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.29.1
- **Preferred DNS:** 8.8.8.8

I would open the **IPv4 properties** of my Wi-Fi adapter and select **“Use the following IP address.”** Then I would enter the above details.

After saving the settings, I would open a browser and check whether the internet is still working.

If the internet works, it means the static IP configuration is working properly.

Important

After completing the test, I would change the setting back to:

Obtain an IP address automatically

This is important because the normal Wi-Fi network uses DHCP to assign IP addresses automatically.

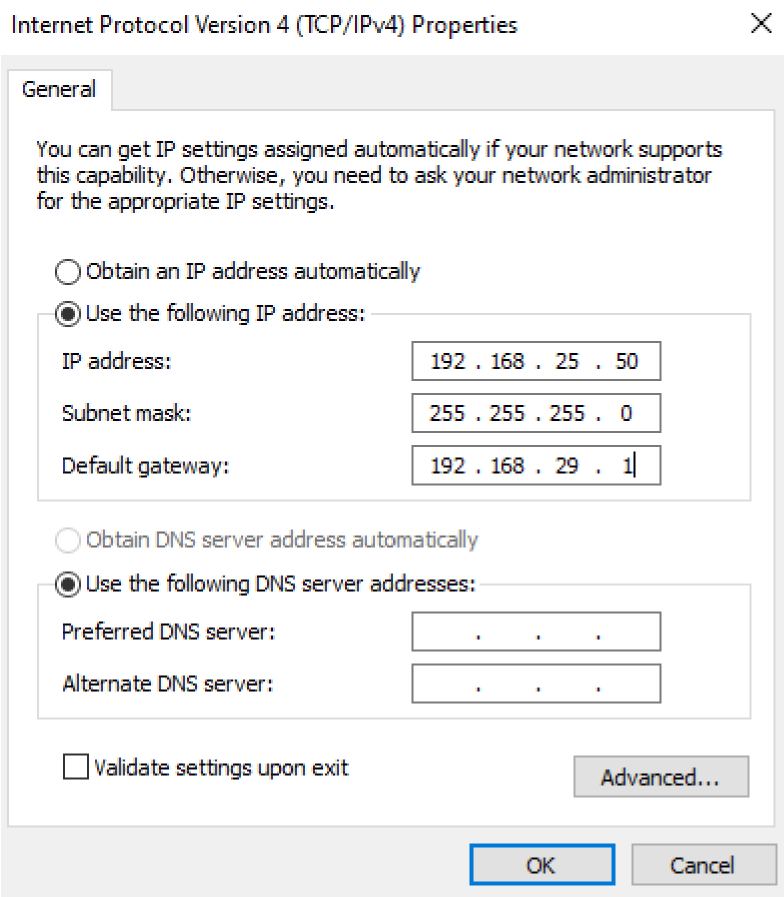

Figure 3: Static IP Address Configuration

Result: This task helped me understand how to manually set a static IP address and keep it in the same network range as the router.

TASK 4 – DHCP RANGE

Practical / Observation Task

In this task, I would open the **Jio router admin panel** and look for the DHCP settings.

The DHCP section normally shows the range of IP addresses that the router can automatically assign to connected devices.

DHCP Range

Current DHCP Range: ____________________________

Write the actual range shown in your Jio router. **Do not make up the range.**

What happens if two devices get the same IP?

If two devices are assigned the same IP address, an **IP address conflict** can occur. The devices may have problems communicating with the network or accessing the internet properly.

Simple Example:

Device A → 192.168.1.20
Device B → 192.168.1.20

Both devices have the same IP, so the network can become confused about which device should receive the data.